

Distance and displacement sensors

http://en.wikipedia.org/wiki/Linear_variable_differential_transformer

Types:

- Inductive, - Capacitive, - Laser light, - Ultrasound, - Radar, - Lidar

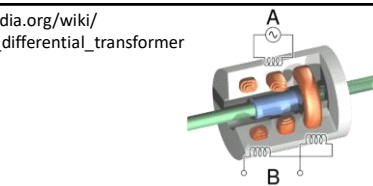
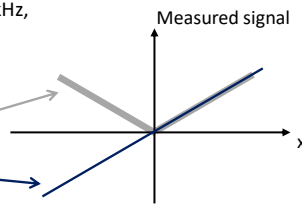
Inductive displacement sensor

(LVDT) Linear variable differential transformer

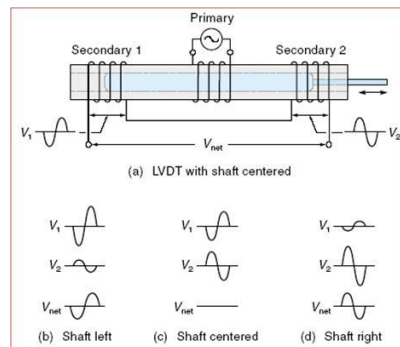
- Configuration: the ac signal applied to the driving coil (A) is detected by placing two oppositely wound coils (B) to the two sides of (A). These "pick-up" coils are connected in series. In the center of the coils there is an iron core, whose position is changed by the object to be measured by means of a rod.
- Principle: Mutual induction varies with the position of the iron core. If the iron core is right in the middle, the pick-up coils are balanced (shaft centered), no signal is detected. If the shaft moves to the left or to the right, a finite signal is detected, whose amplitude linearly increases with the displacement. If just the amplitude is detected, there is no difference between the left and right position, but by lock-in detection (e.g. measuring the Y component with respect to the driving) one obtains a true linear characteristics around zero displacement:
- Typical parameters: 20-20kHz, 1mm- 30cm, mass of the core: ~g, 50-250mV/V/mm

Amplitude measurement with an AC voltmeter

Lock-in detection (Y component)



(Above) Inductive displacement sensor structure
(Below) Signal of the two pick-up coils at different positions of the iron core



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Capacitive displacement sensors

Principle: Capacity (C) depends on the distance (d) between the electrodes, surface area (A) and the dielectric constant of the intermediate region (ϵ_r):

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

Detection may rely on the variation of all these quantities (d, A and ϵ_r), see figures.

Application:

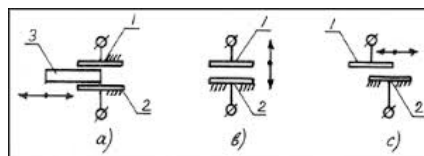
-In semiconductor industry for thickness measurement, positioning for exposure, positioning for hard disk assembly

-Turning machine (lathe): measuring rotational motion roughness and surface smoothness of products

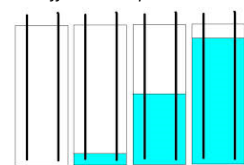
-Etc.

Pros: high accuracy (thickness measurement on <nm scale), can measure in non-contact mode

Cons: requires a clean environment compared to LVDT.



(Above) Different operating principles of capacitive displacement sensors: (a) a dielectric piece is actuated between the electrodes, (b) the electrode distance is varied, (c) the electrodes are moved parallel to each other
(Below) Liquid level measurement: the dielectric constant is different in liquid and in air

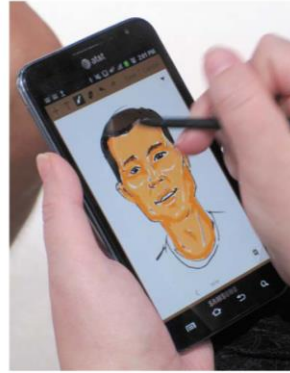


Thin film thickness measurement with nm accuracy, e.g. Alpha-Step IQ Surface Profiler

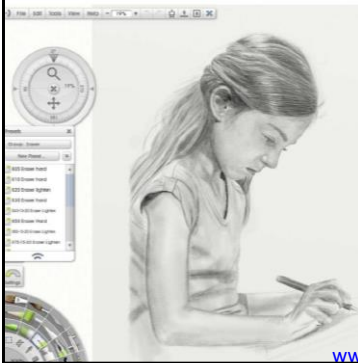
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Touch screen with pressure sensitive pen

Samsung Galaxy Note sketching demo at CES 2012



The Galaxy Notes use both a p-cap touchscreen AND a Wacom EMR stylus (2 sensors!)



Created with an N-Trig active stylus on a Fujitsu Lifebook using ArtRage software

www.walkermobile.com/Touch_Technologies_Tutorial_Latest_Version.pdf (2014)

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Capacitive touch screens

Touch screen based on self-capacity

- 1 armed electrode with capacitance to ground
- Human body capacity increases electrode capacity to ground
- Each electrode is measured separately
- Applied in cheaper devices

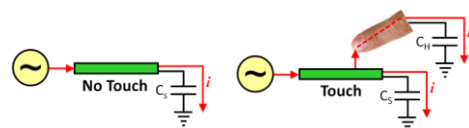
Touch screen based on mutual capacity

- Capacitance between two electrodes
- Human body draws electric field lines, so capacitance decreases
- Each electrode pair crossing is measured separately
- Position accuracy of 0.1-0.2mm can be achieved
- Transparent electrodes are required for touch screens this makes them relatively expensive.

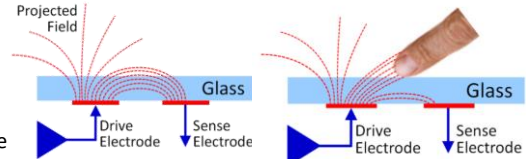
Further sensors for handwriting on a touch screen:

Electromagnetic resonance (EMR) sensor from Wacom.
The pressure on the pen is measured by capacitance variation, 1024 different pressure levels are resolved (see next slide).

Operating principle of touch screens based on self-capacitance



Working principle of touch screens based on mutual capacity



Geometry of a caterpillar-shaped mutual capacitance planar electrode architecture (ELAN)

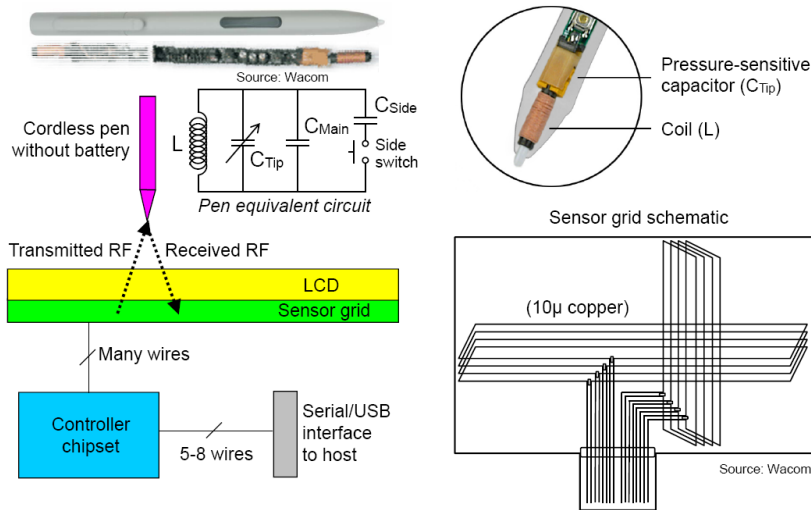


www.walkermobile.com/Touch_Technologies_Tutorial_Latest_Version.pdf (2014)

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Pressure sensitive touch screen pen

www.walkermobile.com/Touch_Technologies_Tutorial_Latest_Version.pdf (2014)



Under the tablet's surface is a printed circuit board with a grid of multiple send/receive coils. In send mode, the tablet generates a electromagnetic field (~531 kHz), which stimulates oscillation in the pen's coil/capacitor (LC) circuit when brought into range of the field. In receive mode, the resonant circuit's oscillations in the pen is detected by the tablet's grid. This information is analyzed by the computer to determine the pen's position, even if the pen does not touch. In addition, the pressure on the pen detunes the resonance frequency of the LC circuit (and thereby the phase of the response), this is the basis of pressure detection. The tablet provides power to the pen through resonant inductive coupling, no battery is required. (Source of text: Wikipedia)

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Laser rangefinder

Amann et al.: Laser ranging: a critical review of usual techniques Optical Engineering, 40 (2001)

Measuring distance with light:

- Interferometric methods (see slides on Metrology): high accuracy
- Time-of-flight (TOF) measurements

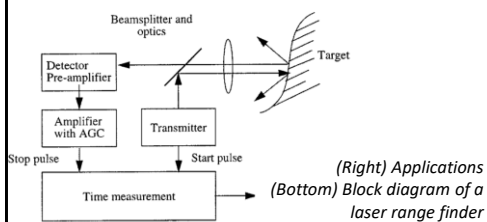
Laser rangefinder (LRF) use the TOF method:

Principle: A laser pulse is sent towards the object, and the time (t_d) it takes for the reflected light to arrive back is measured, $2d = c/t_d$, where d is the distance to the object and c is the speed of light.

Resolution, range: <1mm resolution can be achieved, for this 7ps temporal resolution is required. Can be used at distances of up to several km (e.g. military targets).

Components: Source: 5-50ns laser pulse, Sensor e.g. avalanche photodiode with automatic gain control (AGC). Time measurement unit: triggered by laser pulse and stopped by incoming signal.

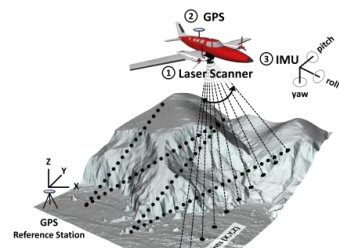
Phase modulation and heterodyne techniques are also applied for laser-based distance measurement (see lab exercise in next semester).



(Right) Applications
(Bottom) Block diagram of a laser range finder



LIDAR (light-radar)
LRF+rotating mirror



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Ultrasonic distance detectors and complex object sensing

Ultrasonic distance detectors

- They work on the principle of sonar: they scan the echo of ultrasound.
- Time-of-flight is the measurement principle here too.
- Source not audible range e.g. 18kHz. A simple microphone is the detector.
- Use in cars : detection of short distances, slow motion
 $v_{\text{sound}}=350\text{m/s}$ $d=30\text{cm} \rightarrow \text{approx. } 1\text{ms}$

Complex distance and environment sensing:

E.g. self-driving cars:

Human error eliminated, 2-3 capacity increase on roads

Car has to deal with a variety of situations $\rightarrow$ many sensors, e.g.:

- parking, accident prevention, distance tracking, lane departure monitoring, blind spot monitoring, cross traffic

Distance sensors: -radar (with radio wave, Doppler shift, fast, good for longer distances, weather independent e.g. for adaptive cruising),

- ultrasonic sensor, with camera recognition
- LIDAR system (light-radar): LRF+rotating mirror (laser, e.g. VelodyneHDL 64 rotating lasers, 1.3 million points/sec $\rightarrow$ virtual model of the environment).

The top diagram shows an ultrasonic sensor emitting a 'Transmit wave' and receiving a 'Reflected wave' from an 'Object' at a certain 'Distance'. Below it, a car is shown with orange ultrasonic waves emanating from its rear. The bottom diagram is a top-down view of a car with various sensor zones: blue for 'RADAR APPLICATION' (lane-change assistance, blind-spot detection, side impact, cross-traffic alert) and yellow for 'ULTRASONIC' (parking assistance/vison, self-parking, lane-departure warning, brake-assistance/collision avoidance, adaptive cruise control).

Figure 2 Several driver-assistance systems are currently using radar technology to provide blind-spot detection, parking assistance, collision avoidance, and other driver aids (courtesy Analog Devices).

<http://www.edn.com/design/automotive/4368069/Automobile-sensors-may-usher-in-self-driving-cars>

Google unveils self-driving car

Google has begun building a fleet of experimental electric-powered cars that will have a stop-go button but no controls, steering wheel or pedals. Google claims that the two-seater vehicle will revolutionise transport by making roads safer, and decrease congestion and pollution

- 1 GPS receiver**
Matches position with customised version of Google's road maps
- 2 Laser range finder:**
Rotating sensor scans 180m distance through 360° to generate 3D map of surroundings
- 3 Video camera**
Identifies other road users, lane markers and traffic signals
- 4 Radars:**
Located at front and rear, detect proximity of obstacles

Speed: Limited to 40km/h to help ensure safety
Engine: 160km-range electric motor – equivalent to one used by Fiat's 500e

Windscreens: Flexible plastic designed to reduce injuries
Front: Foam-like material minimises impact in case of crash
 Car would be summoned with smartphone application

Inertial motion sensors determine velocity and direction

Source and Picture: Google

Lidar system

Velodyne HDL-64E

<http://www.edn.com/design/automotive/4368069/Automobile-sensors-may-usher-in-self-driving-cars>